

Nirmal Nipuna Nanayakkara

Dubai, United Arab Emirates • (+971) 56 363 9349 • nipunann0710@gmail.com

Profile Summary

Senior Software Engineer with 12+ years of experience delivering enterprise web applications and AI-powered business solutions. Strong expertise in Angular, React, TypeScript, Python, Azure AI Foundry, OpenAI, and cloud-native application development. Experienced in designing and integrating LLM-based solutions, RAG, OCR, vector databases, and enterprise APIs to automate business processes and enhance user experiences. Passionate about building scalable AI applications that solve complex real-world problems.

Work Experience – 12+ years

Senior Software Engineer (Angular)

B&S, Dubai, United Arab Emirates — *January 2022 to Present*

- Led the design and development of scalable front-end solutions using Angular, supporting internal enterprise applications across Dubai and the Netherlands for Food, Beauty, Health, and Logistics domains.
- Translated business requirements into wireframes and intuitive UI designs, enhancing user experience while maintaining brand consistency and design standards.
- Defined and implemented front-end architecture, collaborating closely with backend engineers and product managers to deliver robust, scalable solutions for AI applications and optimized application performance by identifying bottlenecks and implementing efficient solutions, resulting in faster processing times.
- Built an intelligent label recognition solution using Azure AI Foundry, Azure OpenAI, OCR, barcode scanning, and .NET APIs to automate warehouse receiving processes.
- Developed mobile applications using Ionic for warehouse operations, streamlining container loading/unloading and significantly improving efficiency and accuracy.
- Contributed to the UI Component Library by developing reusable, scalable components to standardize development across teams.
- Led code reviews, enforced best practices, and contributed to technical documentation to ensure high code quality and maintainability.
- Leveraged AI tools, including Claude AI, to accelerate Angular/TypeScript development, support UI component creation, debug front-end issues, refactor code, and create technical documentation.

Senior Frontend Engineer

Colombo Centre for Cognitive Computing, Colombo, Sri Lanka — *March 2018 - December 2021*

- Created wireframes and translated designs into functional web templates and single-page, scalable applications using Angular and React/Next.js with UI/UX best practices.
- Conducted R&D on next-generation AI interfaces, implementing advanced features in chat applications that improved user experience by 30% using GSAP animation library. Designed and developed intuitive user interfaces for humanoid robots, enabling more natural and engaging human-machine interactions.
- Optimized front-end performance by utilizing browser APIs and minimizing network requests, and worked together with the backend team to ensure smooth integration and implementation of new features.
- Executed comprehensive test cases in Angular, covering Unit Testing, Regression Testing, and End-to-End (E2E) Testing using Jasmine, Karma, and Protractor.
- Managed a team of 3 junior developers, guiding them to enhance their UI skills, and facilitated code reviews and KT sessions.

Software Engineer (Frontend)

Tutor Wizard, Colombo, Sri Lanka — *November 2015 - February 2018*

- Led the design and development phases of an e-learning startup to build the portal and Mobile app using Angular and Ionic Framework, and provided 100,000 students with 24/7 access to course materials and resources, allowing them to learn at their own pace and on their own schedule.
- Worked on developing micro-frontends to improve the application's modularity and scalability, and collaborated with back-end engineers to design and develop a RESTful API for UI integration and communicate data between the applications.

UI/UX Developer

Ben Worldwide (Pvt) Ltd, Colombo, Sri Lanka — *March 2015 - October 2015*

- Created HTML/CSS-based CRM dashboards utilizing jQuery and Bootstrap to support businesses in optimizing their sales processes and monitoring.
- Revitalized WordPress websites and themes, customized plugins, and boosted website security by 40% and SEO scores by 20%.

UI/UX Developer

Dimensions-IT (Pvt) Ltd, Gampaha, Sri Lanka — *August 2012 - March 2015*

- Built responsive/ mobile-first WordPress websites, web applications, and newsletter templates for UK clients to support businesses and optimized websites for SEO, resulting in an additional 20% increase in SEO scores.

Technical Skills

Frontend Technologies

HTML5, CSS3, SCSS, JavaScript ES6+, TypeScript, Angular, Ionic, React, Next.js, React Native, NgRx, Redux, RxJS, GSAP, Three.js, Chart.js, Tailwind CSS, Bootstrap, Material UI, Flexbox, Flutter

Backend Technologies

Node.js, Python, .NET, GraphQL, Firebase, PHP, MySQL, REST APIs, WebSockets, MS Foundry for AI

Other Tools & Platforms

Git, Bitbucket, Grunt, Gulp, NPM, Webpack, CI/CD, Microsoft Azure, AWS, JIRA, Adobe XD, Figma, Claude AI, OpenAI, Codex

Academic Credentials

BSc (Hons) in Computer Science, University College Dublin, Ireland

2010 - 2014

Certifications

Certified Software Engineer – HackerRank hackerrank.com/certificates/be02598e6818

Angular Certification – Udemy udemy.com/certificate/UC-50121f4e-4677-4003-a0eb-93518043fbae/

TestDome Angular Certification (Hard) testdome.com/certificates/aff0860ee6eb45d7a84c88472d42a421

Project Portfolio

Label Recognizer (B&S Logistics) - *Angular, TypeScript, RxJs, .NET, C#, REST APIs, Microsoft Foundry, OpenAI*

- Developed an AI-powered label recognition application for B&S Logistics that enabled warehouse users to capture shipping and logistics labels using tablet devices. Leveraged Microsoft Azure AI Foundry and Azure OpenAI to perform OCR, extract key information from label images, and scan barcodes to automatically identify shipments and validate extracted data.
- Built a responsive application integrated with .NET REST APIs to process captured images, decode barcodes, display extracted fields for user verification, and seamlessly integrate validated information into the logistics system. Automated label processing, reduced manual data entry, improved data accuracy, and accelerated warehouse receiving and shipment handling workflows.

American Red Cross Assistance Chatbot - *React, Next.js, TypeScript, .NET, C#, REST APIs*

- Contributed to the development of the American Red Cross Clara Disaster Assistance Chatbot by building a responsive and accessible web application that streamlined access to disaster relief services and emergency resources.
- Developed scalable Angular components integrated with .NET REST APIs to support conversational workflows, enabling users to locate shelters quickly, access disaster preparedness and recovery information, and navigate available assistance services through an intuitive chat-based interface.

AT&T Summit 2019 Pepper Robot Interactive Assistant - *Angular, TypeScript, RxJs, GSAP, .NET, C#, REST APIs*

- Designed and developed the interactive screen application for the Pepper humanoid robot showcased at the AT&T Summit 2019, enabling attendees to ask questions and receive event-specific information through a conversational interface. Built responsive Angular UI components integrated with backend REST APIs to deliver dynamic content and real-time responses based on summit topics, schedules, exhibitors, and event information.
- Collaborated with backend teams to integrate the application with the Pepper robot, ensuring seamless communication between the tablet interface and the robot during live events while optimizing performance and usability for a high-traffic event environment.

Finastra Power BI Analytics Platform - *React, Next.js, TypeScript, GraphQL, WebSockets, Power BI Embedded*

- Developed a scalable analytics platform using React, Next.js, and Power BI Embedded to deliver interactive dashboards, charts, and real-time business intelligence reports. Integrated GraphQL, WebSocket, and REST APIs to visualize financial data, optimize application performance, and enable data-driven decision-making through responsive and interactive analytics.

Linkedin: linkedin.com/in/nirmal-nipuna-nanayakkara/ • **Github:** github.com/nipunann1991/